Question	Answer	Marks	Partial Marks
2(c)	A: –5, $10 - 3n$ B: $\frac{5}{8}$, $\frac{n}{n+3}$ C: 20, $n^2 - n$	**8**	**B1** for –5 **B2** for $10 - 3n$ oe or **B1** for $k - 3n$ or for $10 - kn$ **B1** for $\frac{5}{8}$ **B1** for $\frac{n}{n+3}$ oe **B1** for 20 **B2** for $n^2 - n$ oe or **B1** for any quadratic or for at least two second differences of 2
3(a)	5	**B1**	
	4	**B1**	
	3.55	**3**	**M2** for (10 × 1 + 6 × 2 + 11 × 3 + 13 × 4 + 14 × 5 + 6 × 6) ÷ 60 oe or **M1** for 10 × 1 + 6 × 2 + 11 × 3 + 13 × 4 + 14 × 5 + 6 × 6 oe
3(b)(i)	42.55 or 42.6	**4**	**M1** for 25, 40, 62.5 soi **M1** for Σfx with x values in correct intervals, including boundaries **M1 dep** on second M1 for $\frac{\Sigma fx}{100}$
3(b)(ii)	10.8 2.16	**2**	**B1** for each or for frequency densities 3.6 and 0.72 seen
4(a)	$\frac{1}{2}(r+5)(r+2)\sin 30 = (r+1)^2$	**M2**	**M1** for $\frac{1}{2}(r+5)(r+2)\sin 30$ oe
	$r^2 + 5r + 2r + 10$ or $r^2 + r + r + 1$ soi	**B1**	
	Leading to $3r^2 + r - 6 = 0$ with no errors or omissions	**A1**	Dependent on both expansions seen

Question	Answer	Marks	Partial Marks
4(b)	$\frac{-1\pm\sqrt{1^2-4(3)(-6)}}{2(3)}$ Or $-\frac{1}{6}\pm\sqrt{2+\left(\frac{1}{6}\right)^2}$ oe or $\frac{1}{3}\left(-\frac{1}{2}\pm\sqrt{18+\left(\frac{1}{2}\right)^2}\right)$ oe	**B2**	**B1** for $\sqrt{1^2-4(3)(-6)}$ or for $\frac{-1+\sqrt{p}}{2(3)}$ or $\frac{-1-\sqrt{p}}{2(3)}$ or $\left(r+\frac{1}{6}\right)^2$ or $\left(3r+\frac{1}{2}\right)^2$
	–1.59 and 1.26	**B1**	
4(c)	9.028 to 9.040	**2**	**M1** for (*their* root (greater than –1) + 1) × 4
5(a)(i)	3 cao	**1**	
5(a)(ii)	–2, –0.45 to –0.4, 2.40 to 2.45	**3**	**B1** each
5(a)(iii)	3 cao	**1**	
5(a)(iv)	Asymptote	**1**	
5(b)(i)	Correct ruled line	**2**	**B1** for ruled line through (0, –2) but not $y = -2$ or for ruled line with gradient 1
5(b)(ii)	1 cao	**1**	
5(c)	Substituting values of x and y into $y = x^2 - \frac{c}{x}$ for an exact point on graph of $y = f(x)$ or substituting *their* value of x from 5b(ii) into $x^2 - \frac{c}{x} = x - 2$	**M1**	
	leading to $c = 2$ with no errors	**A1**	
5(d)	[p =] –1 and [q =] 2 nfww	**2**	**M1** for $x^3 - x^2 + 2x = 2$ seen or **B1** for each nfww
6(a)	4.27 or 4.272...	**2**	**M1** for $4^2 + 1.5^2$ oe